MINING PLAN ALONG WITH PROGRESSIVE QUARRY CLOSURE PLAN FOR CHENDARAPALLI GREY GRANITE QUARRY

(Under Rule 19A of Tamil Nadu Minor Mineral Concession Rules, 1959 and Rule 12, 13 & 16 of Granite Conservation and Development Rules, 1999)
Patta Land/ Lease Period: 20 Years

IN

LOCATION OF THE QUARRY LEASE APPLIED AREA

EXTENT	:	1.80.0 HECTARES
S.F.Nos.	:	223/1A1 (Part) & 224/2A1 (Part)
VILLAGE	:	CHENDARAPALLI
TALUK	:	BARGUR
DISTRICT	:	KRISHNAGIRI
STATE	:	TAMIL NADU

FOR

APPLICANT

M/s. AB Granite,

No.223/A, Chendarapalli Village,
Anchoor Post, Bargur Taluk,
Krishnagiri District, Tamil Nadu – 635 203.
Mobile phone No.: +91 96550 42699 & 90807 81917.
E-mail ID: ab.granite.in@gmail.com.

PREPARED BY

B. Vengadagiri, M.Sc.,

Qualified Person

(Under Rule 15(I)(a) and (b) of MCR 2016)

No.105, 5th Cross, Alagapuram,
Salem District,
Tamil Nadu – 636 010.
Mobile No. +91 86953 32233.
E-mail: vengatb6@gmail.com

ABBREVIATIONS

- EC - Environmental Clearance
- EIA - Environmental Impact Assessment
- EMP - Environment Management Plan
- SEIAA - State Level Environment Impact Assessment Authority
- MoEF&CC - Ministry of Environment, Forest and Climate changes
- MSL - Mean Sea Level
- CPCB - Central Pollution Control Board
- TNPCB - Tamil Nadu Pollution Control Board
- FMB - Field Measurement Book
- S.F.No. - Survey Field Number
- GSI - Geological Survey of India
- DMS - Director of Mines Safety
- DGMS - Director General of Mines Safety
- MMR - Metalliferous Mines Regulations
- MCR - Minerals (Other than Atomic and Hydro Carbons Energy Minerals) Concession Rules
- TNMMCR - Tamil Nadu Minor Mineral Concession Rules
- GCDR - Granite Conservation and Development Rules
- EMP - Environment Management Plan
- NGT - National Green Tribunal
- NONEL - Non Electric
- CRZ - Coastal Regulatory Zone
- HACA - Hill Area Conservation Authority
- QP - Qualified Person
- MT - Metric Tonne
- BGL - Below Ground Level
- CCTV - Closed Circuit Television
- HP - Horse Power
- PSI - Pounds per square inch
- Kva - Kilo-volt-amperes
- m & mm - Meter and millimeter
- ltrs - Liters
- R.L. - Reduced Level

M/s. AB Granite,

No.223/A, Chendarapalli Village,

Anchoor Post, Bargur Taluk,

Krishnagiri District,

Tamil Nadu – 635 203.

Mobile phone No.: +91 96550 42699 & 90807 81917.

CONSENT LETTER FROM APPLICANT

The Mining Plan along with progressive quarry closure plan in respect of Chendarapalli Grey Granite quarry over an extent 1.80.0 Hectares of Patta lands in S.F.Nos. 223/1A1 (Part) and 224/2A1 (Part) of Chendarapalli Village, Bargur Taluk, Krishnagiri District, Tamil Nadu State has been prepared by

Mr. B. Vengadagiri, M.Sc.,
Qualified Person,

We have entrusted the works to prepared the Mining Plan based upon the production requirements to meet as per the Mines Acts, Rules, Regulations and Amendments as on date. We request to the Director, Department of Geology and Mining, Chennai to make further correspondence regarding the corrections of the mining Plan with the said Qualified Person at his following address.

Mr. B. Vengadagiri, M.Sc.,
No.105, 5th Cross,
Alagapuram,
Salem District,
Tamil Nadu – 636 010.
Mobile No. +91 86953 32233.

We hereby undertake that all the responsibilities of contents in the mining plan and if any corrections made in the mining plan by the Qualified Person may be deemed to have been made with our knowledge and consent and shall be acceptable to me and binding on me in all respects. If there is any substantial change during operation, we will carry out a Modified Mining plan and seek its approval from concerned Authorities.

Signature of the applicant
For M/s. AB Granite,

(C. Anandhan)
Partner

Place: Krishnagiri

Date: 14.03.2026

M/s. AB Granite,
No.223/A, Chendarapalli Village,
Anchoor Post, Bargur Taluk,
Krishnagiri District,
Tamil Nadu – 635 203.
Mobile phone No.: +91 96550 42699 & 90807 81917.

DECLARATION OF APPLICANT

The Mining Plan along with progressive quarry closure plan in respect of Chendarapalli Grey Granite quarry over an extent 1.80.0 Hectares of Patta lands in S.F.Nos. 223/1A1 (Part) and 224/2A1 (Part) of Chendarapalli Village, Bargur Taluk, Krishnagiri District, Tamil Nadu State has been prepared in full consultation with me by

Mr. B. Vengadagiri, M.Sc.,
Qualified Person,

I have understood its contents and agree to implement the same in accordance with Laws, Rules and Act applicable to quarry from time to time.

Signature of the applicant
For M/s. AB Granite,

(C. Anandhan)
Partner

Place: Krishnagiri
Date: 18.03.2026

CERTIFICATE FROM THE QUALIFIED PERSON

Certified that I, **B. Vengadagiri, M.Sc.,** residing at No.105, 5th Cross, Alagapuram, Salem District, Tamil Nadu – 636 010 holding a Post Graduate Degree in Geology (M.Sc., Applied Geology) from Annamalai University, Chidambaram and I worked in the field of Mining in a role of Geologist.

Rule 15(I)(a) and (b) of Minerals (Other than Atomic, Hydro Carbons Energy Minerals) Concession Rules 2016 stipulates the eligibility for preparing Mining plans as "(I)(a) a post graduate degree in Geology granted by a university established" and (I)(b) "Professional experience of five years of working in a supervisory capacity in the field of mining after obtaining the degree". Since my qualification and experience are satisfied the Rule (I)(a) and (I)(b) of 15 of the said Rules, I am eligible to prepare Mining Plans for both Major and Minor Minerals.

Accordingly, I prepared this Mining Plan along with Progressive Quarry Closure Plan in respect of Chendarapalli Grey Granite quarry over an extent 1.80.0 Hectares of Patta lands in S.F.Nos. 223/1A1 (Part) and 224/2A1 (Part) of Chendarapalli Village, Bargur Taluk, Krishnagiri District, Tamil Nadu State for **M/s. AB Granite,** having an office at No.223/A, Chendarapalli Village, Anchoor Post, Bargur Taluk, Krishnagiri District, Tamil Nadu – 635 203. Since the Mining Plan is prepared as per the provisions contained in Rule 15(I)(a) and (I)(b) of Minerals (Other than Atomic, Hydro Carbons Energy Minerals) Concession Rules, 2016.

Signature of the Qualified Person

(B. Vengadagiri, M.Sc.,)

Place : Salem
Date : 18.03.2026

B. Vengadagiri, M.Sc.,
No.105, 5th Cross,
Alagapuram,
Salem District,
Tamil Nadu – 636 010.
Mobile No. +91 86953 32233.

CERTIFICATE FROM THE QUALIFIED PERSON

This is to certify that the Provisions of Granite Conservation and Development Rules, 1999 as amended in Tamil Nadu Minor Mineral Concession Rules, 1959 have been observed in the preparation of Mining Plan along with progressive quarry closure plan for Chendarapalli Grey Granite quarry over an extent 1.80.0 Hectares of Patta lands in S.F.Nos. 223/1A1 (Part) and 224/2A1 (Part) of Chendarapalli Village, Bargur Taluk, Krishnagiri District, Tamil Nadu State has been prepared for

M/s. AB Granite,
No.223/A, Chendarapalli Village,
Anchoor Post, Bargur Taluk,
Krishnagiri District,
Tamil Nadu – 635 203.
Mobile phone No.: +91 96550 42699 & 90807 81917.

Whenever specific permissions/exemptions/ relaxations and approvals are required, the applicant will approach the concerned authorities of Director of Geology and Mining, Government of Tamil Nadu, Guindy, Chennai– 600 032 for such permissions/ exemptions/ relaxations and approvals.

It is also certified that information furnished in the above Mining plan are true and correct to the best of my knowledge.

Signature of the Qualified Person

(B. Vengadagiri, M.Sc.,)

Place : Salem
Date : 18.03.2026

B. Vengadagiri, M.Sc.,
No.105, 5th Cross,
Alagapuram,
Salem District,
Tamil Nadu – 636 010.
Mobile No. +91 86953 32233.

CERTIFICATE FROM THE QUALIFIED PERSON

Certified that the Provisions of Mines Act, Rules and Regulations made there under have been observed in the preparation of Mining Plan along with progressive quarry closure plan for Chendarapalli Grey Granite quarry over an extent 1.80.0 Hectares of Patta lands in S.F.Nos. 223/1A1 (Part) and 224/2A1 (Part) of Chendarapalli Village, Bargur Taluk, Krishnagiri District, Tamil Nadu State has been prepared for

M/s. AB Granite,
No.223/A, Chendarapalli Village,
Anchoor Post, Bargur Taluk,
Krishnagiri District,
Tamil Nadu – 635 203.
Mobile phone No.: +91 96550 42699 & 90807 81917.

Whenever specific permissions/exemptions/ relaxations and approvals are required, the applicant will approach the concerned authorities of the Director of Mines Safety, No. 5, IInd Street, Block – AA, Anna Nagar, Chennai, Tamil Nadu for such permissions/ exemptions /relaxations and approvals.

It is also certified that information furnished in the mining plan are true and correct to the best of my knowledge.

Signature of the Qualified Person

(B. Vengadagiri, M.Sc.,)

Place : Salem
Date : 18.03.2026.

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MINING PLAN ALONG WITH PROGRESSIVE QUARRY CLOSURE PLAN FOR CHENDARAPALLI GREY GRANITE QUARRY

(Under Rule 19A of Tamil Nadu Minor Mineral Concession Rules, 1959 and 12, 13 and 16 of Granite Conservation and Development Rules, 1999)

INTRODUCTION AND EXECUTIVE SUMMARY:

The company **M/s. AB Granite,** having an office at No.223/A, Chendarapalli Village, Anchoor Post, Bargur Taluk, Krishnagiri District, Tamil Nadu – 635 203 has entrust and given consent to preparation of Mining plan and Progressive Mine Closure Plan as per the provisions of Mines Act, Rules, Regulations and as amended till date.

The company has applied quarry lease for quarrying Grey Granite for over an extent 1.80.0 Hectares of Patta lands in S.F.Nos. 223/1A1 (Part) and 224/2A1 (Part) of Chendarapalli Village, Bargur Taluk, Krishnagiri District, Tamil Nadu State for a period of twenty years under Rule 19-A of Tamil Nadu Minor Mineral Concession Rules, 1959.

The application was examined and processed by the Natural Resources (MMB.2) Department, Secretariat, Government of Tamil Nadu and passed a Precise Area Communication letter vide **letter No. 3187/MMB.2/2025–1, Dated: 14.03.2026** under sub-rule (13) of Rule 19-A of Tamil Nadu Minor Mineral Concession Rules, 1959 for submission of approved Mining plan within 90 days and obtain Environmental Clearance from the State Level Environment Impact Assessment Authority, Tamil Nadu under rule 42 of Tamil Nadu Minor Mineral Concession Rules, 1959 (Refer Annexure No. I).

This Mining Plan along with Progressive quarry closure Plan is prepared in full consent with **M/s. AB Granite,** Krishnagiri District in respect of Chendarapalli Grey Granite quarry over an extent **1.80.0 Hectares of Patta lands in S.F.Nos. 223/1A1 (Part) and 224/2A1 (Part) of Chendarapalli Village, Bargur Taluk, Krishnagiri District**. This mining plan is prepared under Rules 19-A and 42 of Tamil Nadu Minor Mineral Concession Rules, 1959 and Rules 12, 13 and 16 of Granite Conservation and Development Rules, 1999 with obtained full consent as per the application and Production schedule in preparation of Mining plan as per the provisions of Mines Act, Rules, Regulations as on date.

This Mining plan has been prepared after carrying the field investigation, geological survey, collection of Primary & secondary data, geological attributes, environmental setting and estimated the Resources & Reserves, depth of mining as identified in the field with best our knowledge and experience. This mining plan is prepared by considering as amended Rule 42 of Tamil Nadu Minor Mineral Concession Rules, 1959 and as per the EIA Notification 2006 and its subsequent Amendments.

This mining plan has been prepared by keeping and considering all the parameters stipulated by the Government of Tamil Nadu before and during the course of quarry operations.

The company ensures to comply all the condition stipulated by the Government before the execution of lease deed and during the course of quarrying operations.

In order to ensure compliance of the order of the Honorable Supreme Court Dated: 27.02.2012 in I.A.No.12-13 of 2011 in Special Leave Petition (SLP) (C) No 19628-19629/2009, it has been now decided that all mining projects of minor minerals including their renewal is require prior environmental clearance. As per amendment in EIA Notification 2006 vide S.O. 1886(E), Dated:20.04.2022 "All mining lease area in respect of minor mineral mining leases and ≤ 250 ha mining lease area in respect of major mineral mining lease other than coal" would be treated as category B and will be considered by the state notified by Ministry of Environment, Forest and Climate Change as prescribed procedure under EIA notification 2006.

The field investigation was carried out by the Qualified Person and his Team as on 16.03.2026.

1.0 GENERAL:

1.1 Name of the applicant with address:

Name	:	**M/s. AB Granite,**
Address	:	No.223/A, Chendarapalli Village, Anchoor Post, Bargur Taluk, Krishnagiri District
State	:	Tamil Nadu
Pin code	:	635 203.
Phone	:	+91 96550 42699 & 90807 81917.
E-mail ID	:	ab.granite.in@gmail.com.
Aadhaar No.	:	9243 2869 9729.

1.2. Status of the applicant:

M/s. AB Granite is a partnership firm and the Partnership deed executed on 22.01.2024 under the Indian Partnership act, 1932 with two partners. The details of partners are given table below (Refer annexure Nos. VIII).

Table – 1

S.No.	Name	Address	Designation
1	Mr. C. Anandhan, S/o. Chinnappan	No. 226A, Modikuppam, Nakkalpatti, Krishnagiri, Tamil Nadu – 635 203.	Partner
2	Mr. G.Subramani, S/o. Govindhasamy	No. 6/227, Gandhi Nagar, Modikuppam, Anchoor, Krishnagiri, Tamil Nadu – 635 203.	Partner

Mr. C. Anandhan is the partner and he is an authorized person for signing all the documents on behalf of the firm (Refer annexure Nos. IX).

1.3. Mineral which the applicant intends to mine:

The applicant intends to quarry **Grey Granite** dimensional stone and Petrologicaly called as **Granitic gneiss.**

1.4. Period for which the mining area is required:

Twenty years as mentioned in Precise area Communication letter.

1.5. Name and address of the Qualified Person who prepared the mining plan:

Name	:	B. Vengadagiri, M.Sc., Qualified Person (As per Rule 15(I)(a) and (b) of MCR 2016)
Address	:	No.105, 5th Cross Alagapuram Salem District Tamil Nadu – 636 010
Mobile No.	:	+91 86953 32233
E-mail ID	:	vengatb6@gmail.com

1.6. Name and address of the prospecting agency:

State Geology and Mining Deptartment, Govt. of Tamil Nadu, has carried out the Regional prospecting and exploration in these areas during 1992 to 1993.

Geological Survey of India has carried out detailed mapping of the commercial granite deposits of Tamil Nadu. Besides, the Qualified Person and his team members made a detailed geological study of the area and demarcated clearly the Grey granite deposit with a mine surveyor. The granite formation is clearly visible right from the nearby existing pits. No detailed prospecting carried out by any agencies.

Address of the prospecting Agency:

(i) STATE GEOLOGICAL DEPARTMENT
O/o The Director of Geology and Mining
Thiru Ve Ka industrial Estate,
Guindy, Chennai – 32.

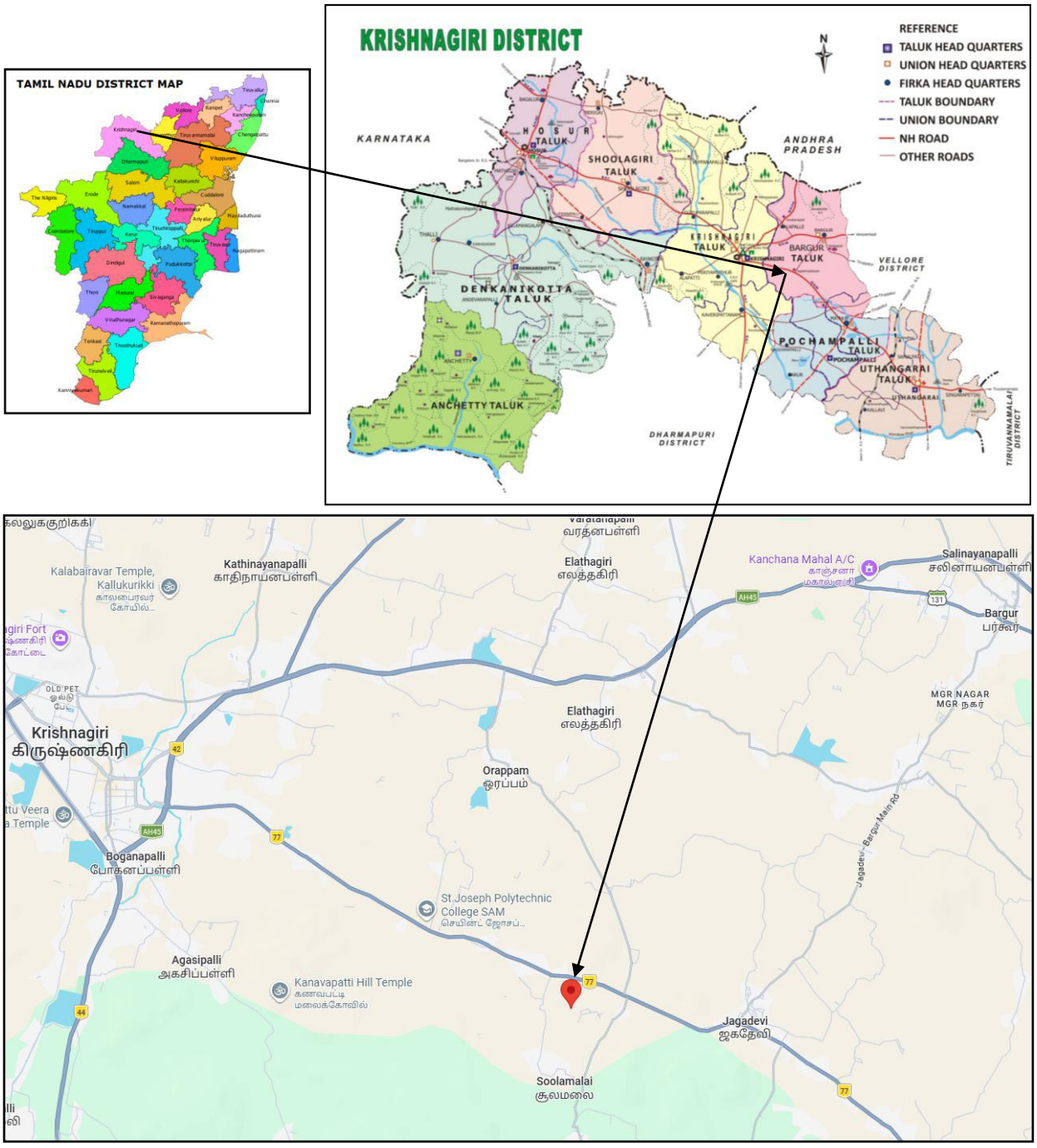
1.7. Reference no. Date of grant/ Precise Area Communication received form the State Government:

The Precise area Communication letter has been issued by the Natural Resources (MMB.2) Department, Secretariat, Government of Tamil Nadu and passed a Precise Area Communication letter vide **letter No. 3187/MMB.2/2025–1, Dated: 14.03.2026** (Refer Annexure No. I).

2.0 LOCATION AND ACCESSIBILITY:

The lease applied area is located about 8km Southeastern side of Krishnagiri city, 9.5km Southwestern side of Bargur town and 1km Western side of Chendarapalli hamlets.

Location Map of the Lease Applied area



2.1 Details of the area:

Table – 2

District and State	Taluk	Village	S.F.No.	Area (Ha.)	Patta No.	Classification
Krishnagiri and Tamil Nadu	Bargur	Chendarapalli	223/1A1(P)	1.38.0	1454	It is Patta Land, classified as Punsei (Refer Annexure Nos. IV - VI).
			224/2A1 (P)	0.42.0	3409	
			Total	**1.80.0**		

Source: As per the FMB, Patta and 'A' register record furnished by the applicant.

2.2 Whether the area recorded to be in Forest Department:

The area does not falls under forest land of any category. It is patta lands.

2.3 Ownership / Occupancy and Classification of the area:

The applied area is patta lands, classified as punsei. The lands are registered in the name of Thiru. G.Subramani (Partner of the company) Vide Patta No.1454 and 3409 (Please refer Annexure No. IV to VI). The company has obtained lease deed from the pattadar for a period of 35 years with effect from execution of lease deed (Please refer Annexure No. VII).

2.4 Existence of public road / Railway line, if any nearby and approximate distance:

The approach road is available on the western side, which is leads to National Highway (NH77) road located at 570m on the Northwestern side of the area. There is no other patta lands are encountered for the haulage of Grey Granite (Please refer Plate No.I to IC).

Road access is available from the quarry to state highways and National Highway, the traffic density is not much more due to the transportation of quarried out material.

The Nearest Railway line is Salem - Jolarpet which is located about 25km on the Southwestern side of the area as per the GSI Toposheet and Google Map.

The nearest town is Krishnagiri which is located about 8km on the Northwestern side of the area, where basic facilities like Hospital, Communication centre, Schools, Police Station and Bus terminus are available. District head quarters and District Administrative Offices are available in Krishnagiri located at 8km on the Northwestern side of the area.

2.5 Toposheet No. with Latitude and Longitude:

The area is marked in the Survey of India, Topo Sheet No. 57L/07. The area lies between the Latitudes of 12°29'04.19"N to 12°29'11.96"N and East Longitudes of 78°17'27.71"E to 78°17'33.17"E on WGS datum-1984. (Plate No. I & II) (Please refer Plate Nos. I to II as per the GSI Toposheet).

3.0 GEOLOGY AND EXPLORATION:

3.1 Topography:

The area is situated in flat terrain. The gradient is gentle towards Southern side and altitude of the area is 498m above from MSL. The Grey granite is clearly visible right from the existing quarry pit on the Eastern side adjacent to the area and other places are concealed under Reddish gravelly soil having an average thickness of 2m and 4m Weathered rock. The Grey Granite is medium to coarse grained with feldspar and quartz is major constituents and Pyroxene, Garnet, Mica and other mafic minerals are accessories. This gneissic formation is having wavy pattern of alternate layer of light and dark colour minerals which adds the austenitic beauty for this rock.

3.2 Regional Geology & Geological Succession:

The Grey Granite is medium to coarse grained with feldspar and quartz is major constituents and Pyroxene, Garnet, Mica and other mafic minerals are accessories. The petrological settings of the area are simple and not a complicated phenomena. There are no major minerals observed in the vicinity of the proposed quarry. A brief description of the regional Geology is discussed below.

3.2.1 Geology of the Krishnagiri District:

This Grey granite is Petrologically called as **"Granitic Gneiss"** which is widely used for slabs, Tiles and Mounments after cutting and polishing.

The Krishnagiri district is underlain by hard Crystalline rocks of Archaean age comprising of various rock types such as Gneiss, Charnockite, etc.,. The Gneissic type of Crystalline formation is found in the North and Northeastern part of the District. Shoolagiri, Hosur, mattur and soolamalai areas covered by Granitic Gneiss (Migmatite). The Late Archean crust of Krishnagiri, Tamil Nadu consists of tonalitic-trondhjemitic-granodioritic (TTG) gneisses with mafic and sedimentary enclaves, formed between 2.7 and 2.5 Ga and metamorphosed at amphibolite facies in the north to granulite facies in the south close to 2.5 Ga. Migmatization occurred at all grades, and numerous small granite bodies were emplaced near the amphibolite-to-granulite facies horizon. The Grey granites are seen to the central part and the SE part of the district, more specifically in Rayakottai, Kaveripattinam, Jagadevi and Velampatti. These dimensional blocks are quarried to make a polished stone, slabs, monuments etc.,

Granites were formed from molten rock referred to as "Magma" formed at great depths within the crust of the earth. Such rocks that were formed at great depths during the Archaean age are now exposed at the surface of the earth as a result of the combined actions of wind, air, sun and water and weathering and denudation over the past several million years.

LITHOSTRATIGRAPHY OF KRISHNAGIRI DISTRICT:

The general geological sequence of the rock types in the area is:-

Order of super position:-

ROCK TYPE	AGE
Soil	Pleistocene to Recent
---------- Unconformity ----------	
Dolerite dykes	Archaean to Proterozoic
Migmatites (PGC-II)	
Granites	
Charnockite group	
Peninsular Gneissic Complex- I	

3.2.2. Geology of the lease applied area:

The The quarry lease area contains Granite gneiss. The grey granite deposit is mostly concealed under soil with an average thickness of 2m and 4m weathered rock and followed by fresh grey granite. The grey granite is clearly visible from the existing quarry pit. The rock formation is popularly known as Granitic gneiss essentially made up of a supra crustal assemblages of quartz and Feldspar and Pyroxene, Mica, Garnet and other mafic minerals are accessories. The lease applied area comprises Granitic gneiss popularly termed as **"Paradiso"**.

The rock type is euhedral, medium to coarse grained, inequigranular, bulk density is 2.65 and well developed gneissic banding of alternate layers of light and dark colour minerals are the specialty of this area which denotes the indicative of flow pattern of the rock mass in N20^{o}W- S20^{o}E direction (i.e., the cutting direction of the grey granite). The colour of the rock is grey which may find a good market for dimensional granite stones.

Some slender pegmatite veins are intruded in a crisscross fashion and well developed strike and dip joints and xenoliths are observed at the surface level which is likely to decrease in deep seated condition. Taking in to consideration of the above geological factors, over burden, inter burden wastage during quarrying, other flaw and flower patches etc., an average recovery percentage has been estimated as 40% upto 26m (2m Topsoil + 4m weathered rock + 20m grey granite) depth.

The Physical attitude of the Grey Granite deposit of this area is given below:-

Strike Direction	-	N20^{o}W - S20^{o}E
Dip amount and direction	-	Almost Vertical.

3.3 Details of Exploration:

3.3.1. Already Carried Out:

The Grey granite is clearly visible from the existing quarry pit on the Eastern side adjacent to the area. As far as Grey Granite deposits are concerned, the only practical method is the systematic geological mapping and delineation of commercial Grey Granite bodies within the field and careful evaluation of body luster, physical properties, engineering properties, commercial aspects etc.

Such an exploration study has already been conducted regionally in this area by the Geological Survey of India (GSI) in the year 1966 and Department of Geology and Mining of Tamil Nadu in year 1992 to 1993.

Based on the valuable geological information and by the field investigation carried out by the QP and his team, the estimation of geological resources, mineable reserve is arrived and assumed based upon the anticipated waste saleable granite blocks and present market scenario.

3.3.2 Proposed study to be carried out:

From the Petrogenetic character of the rock the depth persistence of the Grey Granite stone may beyond the workable limit, only 26m (2m Topsoil + 4m Weathered rock + 20m Grey Granite) depth persistent has been taken as economically viable (at present scenario considering for the entire lease Period and not for the entire life of the mine) to estimate categories of proved, probable and possible reserves.

The recovery of saleable Grey Granite stones has been estimated as 40% and if the recovery percentage is good it may enhance or or bad it may decrease respectively. This estimation is based upon the machenaries available in the market at present scenario and the market demand for this type of granite.

No definite programs for future exploration have been proposed. The quarrying activities for the next five years with deep cut as envisaged in the mining plan may render additional data as may be required for future planning. The attributes of the geological formation with repect to the saleable granite and waste ratio will be discussed in the ensuing scheme period.

3.4 Method of Estimation of Reserves:

The Geological plan demarcating the commercially marketable granite body has been prepared in 1:1000 Scale, totally three sections have been drawn, one cross section is drawn length wise as (X-Y) and other two cross sections are drawn Width wise as (A-B and C-D), Which are suitably chosen to cover the maximum area, in the scale of 1:1000 (Refer Plate No. IV).

The cross sectional area for the proved depth persistence of 26m for each section. The cross sectional area multiplied by its length x breadth x Depth gives the volume (insitu) in the area wise. The sum total of the insitu reserves available within the block gives the geological resources of the quarry lease applied area.

From the total geological insitu resources, the quantity of saleable granite stones, quantity of rejects and waste generation are estimated by applying recovery factor as 40% by its volume. High efficient technology machineries, quarry masters, Market demand significantly determine the recovery percentage of granite quarries. The estimated recovery is based on today market scenario and available technology. This estimated recovery has been considered as only normative recovery. When the market demands, the applicant may take necessary steps to deploy a quarry masters with latest innovative machineries technology. So the recovery enhancement may raise to the peak production resulting more than 40%. During the operation the method of quarry, deployment of men and machineries will not have any significant impact on the Environment. It is worth mentioning the recovery anticipate the normative production has been scientifically converted into commercial production resulting in the decrease dump of waste inside the quarry. Due to the micro fractures, flaws, patches, xenoliths, required dimension, dressing, etc., the recovery in the granite may not be 100% of the R.O.M. As the sale of Grey Granite stone are in terms of Metric Tonne as in the case of major industrial mineral, the geological resources, Mineable reserves and quantity of waste generated etc are given in terms of Metric Tonne. The bulk density of Grey Granite is 2.65 and 2 for Topsoil and Weathered rock has been considered for estimation of reserves.

The details of estimation of geological resources and mineable reserves with reference to the geological plan & cross sections and conceptual plan and sections as shown in Plate No. IV and IX respectively has been furnished.

3.5 Estimation of Geological Resources and Grade:

Maximum Length = 209m
Maximum Width = 93m
Maximum Depth = 26m (6m Over burden + 20m Granite)

Due to uneven shape of the area, the geological resources are estimated by area method and the resources are as follows:

Table – 3

GEOLOGICAL RESOURCES

Area (Ha)	Area/Ha (m^2)	Area (m^2)	Depth (m)	Volume (m^3)	Bulk Density	ROM (MT)	Granite Recovery @ 40% (MT)	Mineral Reject @ 60% (MT)	Weathered Rock (MT)	Topsoil (MT)
1.80.0	10000	18000	2	36000	2	-	-	-	-	72000
			4	72000	2	-	-	-	144000	-
			20	360000	2.65	954000	381600	572400	-	-
Total				**468000**		**954000**	**381600**	**572400**	**144000**	**72000**

Total Geological Resources of ROM = 9,54,000 MT
Total Recoverable Resources @ 40% = 3,81,600 MT
Granite waste @ 60% = 5,72,400 MT
Weathered rock = 1,44,000 MT
Total Waste (Granite waste + Weathered rock) = 7,16,400 MT
Topsoil = 72,000 MT
Granite : Waste ratio = 1:1.87

The Geological resources estimated based on the geological cross sections up to the economically workable depth of 26m below from the ground profile at the rate of 40% recovery yields 3,81,600 MT and 9,54,000 MT of ROM. Geological Resources has been estimated based on the physical investigation and filed survey data.

3.6 Tentative Mineable Reserves and Grade:

Maximum Length = 193m

Maximum Width = 76m

Maximum Depth = 26m (6m Over burden + 20m Granite)

Table – 4

MINEABLE RESERVES											
Section	**Bench**	**Length (m)**	**Width (m)**	**Depth (m)**	**Volume (m^3)**	**Bulk Density**	**ROM (MT)**	**Granite Recovery @ 40% (MT)**	**Mineral Reject @ 60% (MT)**	**Weathered Rock (MT)**	**Topsoil (MT)**
XY-AB	i	54	76	2	8208	2	-	-	-	-	16416
	ii	50	69	4	13800	2	-	-	-	27600	-
	iii	46	61	5	14030	2.65	37179.50	14871.80	22307.70	-	-
	iv	41	51	5	10455	2.65	27705.75	11082.30	16623.45	-	-
	v	36	41	5	7380	2.65	19557.00	7822.80	11734.20	-	-
	vi	31	25	5	3875	2.65	10268.75	4107.50	6161.25	-	-
	Total				**57748**		**94711.00**	**37884.40**	**56826.60**	**27600**	**16416**
XY-CD	i	139	66	2	18348	2	-	-	-	-	36696
	ii	135	60	4	32400	2	-	-	-	64800	-
	iii	131	52	5	34060	2.65	90259.00	36103.60	54155.40	-	-
	iv	126	41	5	25830	2.65	68449.00	27379.60	41069.40	-	-
	v	121	31	5	18755	2.65	49701.00	19880.40	29820.60	-	-
	vi	116	21	5	12180	2.65	32277.00	12911.00	19366.00	-	-
	Total				**141573**		**240686.00**	**96274.60**	**144411.40**	**64800**	**36696**
Grand Total					**199321**		**335397.00**	**134159.00**	**201238.00**	**92400**	**53112**

Total Mineable Reserves ROM = 3,35,397 MT

Tentative Mineable Reserves @ 40% = 1,34,159 MT

Granite waste @ 60% = 2,01,238 MT

Weathered rock = 92,400 MT

Total Waste (Granite waste + Weathered rock) = 2,93,638 MT

Topsoil = 53,112 MT

Granite : Waste ratio = 1:2.19

Mineable reserves have been estimated as 1,34,159 MT at the rate of 40% recovery and 3,35,397 MT of ROM upto a depth of 26m from the existing ground profile. The mineable reserves are estimated by deducting the mineral locked up area under safety distance and bench loss. Hence, the remaining area is taken for estimation of mineable reserves.

The Grey Granite body occurring in this area exhibits more or less uniform colour and texture. If any variation occurs during quarrying, such as cracks, joints, patches, colour variations etc, the defective area will be removed by chiseling. The formation is more or less uniform and no gradational change is noticed except some patches, shears and cracks.

4.0 MINING:

Open cast semi mechanized mining with 5.0 meter vertical bench with a bench width of 5.0 meter is being proposed.

Under the regulation 106 (2) (a) of the Metalliferous Mines Regulation 1961, in all open cast mining, the bench height should not exceed, 6.0 meter and bench width should not be less than bench height of the vertically cut face.

But as far as the mining of granite dimension stones are concerned, observance of the provisions of Regulation 106(2) (a) and (b) is available with Director General of Mines Safety. If the applicant intends to modify the dimensions of benches, relaxation and permission are available with Director General of Mines Safety under 106 (2) (a) and (b) of Metalliferous Mines Regulations, 1961. If any relaxation is required the applicant may apply to the Director of Mines Safety for which the provisions of relaxation of working on benches is possible as per the Regulation of Mines and Minerals (Development and Regulation) Act, 1957. This relaxation will be applied and obtained after the execution of lease/Commencement of quarry operation. In such a scenario if there is any drastic change in the Resources and Reserves a modified plan will be submitted to the concerned authority for necessary relaxation, clearance and permission.

The production of Grey Granite dimension stone in this quarry involves the following method which is typical for granite stone quarrying in contrast to other major mineral mining. Splitting of rock body of considerable volume from the parent rock formation by carefully avoiding visibly seen defects such as patches veins, etc., is done by adopting the method of "diamond wire cutting" along the horizontal as well as two vertical sides on the front face of the formation. This liberation of huge volume of granite body from the parent sheet rock is called "primary cutting". This huge portion is further split in to several blocks of desirable dimensions. The blocks thus splitted are removed from the pit to the dressing yard by using Crawler crane for further dressing.

Removing the defective portions and dressing them in to the dimension blocks are done manually using feather and wedges and chiseling respectively by the experienced skilled labours or by innovative machineries. The defect free, dimensional stone of different sizes as approved in the market are thus produced by the method as described above, and the process is continuously monitored by applicant's experienced quality control personnel.

The waste material generated during quarrying activity includes rock fragments of different sizes, and also during dressing of the blocks. As and when there is accumulation of waste, the same is loaded into the tipper by loading machines and dumped in the respective places ear-marked for the purpose (Plate. No. VI). The quarried out topsoil will be preserved all along the safety zone and utilized for bund formation and green belt purpose.

4.1 Year Wise Development and Production for the First Five Years:

Total Length = 70m

Maximum Width = 69m

Maximum Depth = 26m (2m Topsoil + 4m Weathered rock +20m Granite)

Table – 5

YEARWISE RESERVES												
Section	**Year**	**Bench**	**Length (m)**	**Width (m)**	**Depth (m)**	**Volume (m^3)**	**Bulk Density**	**ROM (MT)**	**Granite Recovery @ 40% (MT)**	**Mineral Reject @ 60% (MT)**	**Weathered Rock (MT)**	**Topsoil (MT)**
XY-CD	I	i	31	66	2	4092	2	-	-	-	-	8184
		ii	26	60	4	6240	2	-	-	-	12480	-
		iii	18	52	5	4680	2.65	12402.00	4960.80	7441.20	-	-
		iv	8	41	5	1640	2.65	4346.00	1738.40	2607.60	-	-
		Total				**16652**		**16748.00**	**6699.20**	**10048.80**	**12480**	**8184**
	II	i	17	66	2	2244	2	-	-	-	-	4488
		ii	17	60	4	4080	2	-	-	-	8160	-
		iii	14	52	5	3640	2.65	9646.00	3858.40	5787.60	-	-
		iv	14	41	5	2870	2.65	7605.50	3042.20	4563.30	-	-
		Total				**12834**		**17251.50**	**6900.60**	**10350.90**	**8160**	**4488**
	III	i	20	66	2	2640	2	-	-	-	-	5280
		ii	20	60	4	4800	2	-	-	-	9600	-
		iii	15	52	5	3900	2.65	10335.00	4134.00	6201.00	-	-
		iv	15	41	5	3075	2.65	8149.00	3260.00	4889.00	-	-
		Total				**14415**		**18484.00**	**7394.00**	**11090.00**	**9600**	**5280**
	IV	iii	8	52	5	2080	2.65	5512.00	2204.80	3307.20	-	-
		iv	8	41	5	1640	2.65	4346.00	1738.40	2607.60	-	-
		v	16	31	5	2480	2.65	6572.00	2628.80	3943.20	-	-
		Total				**6200**		**16430.00**	**6572.00**	**9858.00**		
	V	v	19	31	5	2945	2.65	7804.25	3121.70	4682.55	-	-
		vi	25	21	5	2625	2.65	6956.25	2782.50	4173.75	-	-
		Total				**5570**		**14760.50**	**5904.20**	**8856.30**		
	Grand Total					**55671**		**83674.00**	**33470.00**	**50204.00**	**30240**	**17952**

Total Proposed Reserves ROM	=	83,674 MT
Total proposed Reserves @ 40%	=	33,470 MT
Granite waste @ 60%	=	50,204 MT
Weathered rock	=	30,240 MT
Total Waste (Granite waste + Weathered rock)	=	80,444 MT
Topsoil	=	17,952 MT
Granite : Waste ratio	=	1:2.4

The average annual production per year would be 6,694 MT for the first five year plan period considering at the rate of 40% recovery. **Annual peak production capacity is 7,394 MT @ 40% recovery in 18,484 MT of ROM**. The proposed production schedule is arrived as per requirement by the company and the EIA and EMP will be prepared for an annual peak production.

The proposed year wise quantum of excavation and the details of estimation of production quantity and generation of waste are furnished with reference to the year wise development and production plan (Plate No.V) More details of the year wise production parameters are explained with bench length, width and height in Plate No. V.

4.2 Proposed rate of production when the mine is fully developed:

The proposed rate of production where the quarry is fully developed is 6,694 MT per annum @ 40% recovery. The production schedule in the subsequent five years are drawn mainly in consideration of reserves position, market demand and the cost of production.

4.3 Mineable reserves and anticipated life of quarry:

The depth persistence of the formation will be beyond the economically workable depth. The method of extraction from the sheet rock is highly expensive affair at greater depths at present scenario.

An optimum depth of 26m depth has been proposed as economically viable depth. Eventually this depth is the optimum for safe and scientific quarrying.

The mineable reserves are calculated by excluding the quarry loss due to formation of benches with suitable height & width upto ultimate depth of quarry and the mineral reserve held up within the safety distance all along the area boundary.

Estimated Life of Quarry

Mineable Recoverable Reserves @ 40%	= 1,34,159 MT
Average Production per Year @ 40%	= 33,470 MT/5 years = 6,694 MT
Estimated Life of the Quarry	= 1,34,159 MT/6,694 MT
	= 20 Years.

Based on the above, and taking into consideration of the available Mineable reserves, **the life of quarry is anticipated about 20 years** (considering all the safety factors) at 40% recovery, if the quarry is being worked continuously with an average annual production of 6,694 MT. This calculation is based on the plan approved by Director of Mines Safety leaving Benches and Safety barriers. If the annual production increases considerably and consistently a modified mining plan will be prepared under Granite Conservation and Development Rules, 1999 the same will be submitted to the relevant authorities for subsequent clearance and approval.

4.3.1 CONCEPTUAL MINING PLAN

Conceptual mining plan is prepared with an object of long term systematic development of benches; lay outs, selection of permanent ultimate pit limit, depth of quarrying and ultimate pit, selection of sites for construction of infrastructure, etc.,

The ultimate pit size is designed based on certain practical parameters such as economical depth of quarrying, safety zones, permissible area, etc.,

The ultimate pit dimension of the quarry is given below.

ULTIMATE PIT DIMENSIONS

Table – 6

Maximum Dimensions in meters			
Section	Length	Width	Depth
XY-AB	54	76	26m (R.L. 498m – R.L. 472m)
XY-CD	139	66	26m (R.L. 498m – R.L. 472m)

However, during extraction of blocks each bench will be of 5m height with vertical slope for proper dimension cutting. The quantity of excavation is estimated to be 4,80,909 MT (ROM 3,35,397 MT + Weathered rock 92,400 MT + Topsoil 53,112 MT) to a depth of 26m. The generation of total waste is estimated about 2,93,638 MT and marketable Grey Granite as 1,34,159 MT. The Conceptual Mining is based upon the entire ROM proposed for the life of the Mine.

During the first five years the excavated waste (80,444 MT) will be proposed to dump on the Northern side with maximum geometry of (Area) 1950m x (Height) 17.46m, which will act as temporary waste dump. After expiry of the lease period if the mineral reserves available and Market persist, the applicant may apply a renewal of quarry lease as to develop and conserve mineral reserves. If the applicant attains the permission form the competent authiorities to dump the waste in the non mining zone the attributes of the waste dump will be disussed in the ensuing scheme period.

As per G.O.(Ms)No.04, Natural Resources (MMC.1) Department, Dated: 22.08.2023, if permission is granted for removal of waste from concerned authorities, the waste material will be supplied to the needy crusher to making the M-Sand, building and road construction material after paying the seniorage fee and obtained necessary clearance and approval from concerned department for handling the waste. When the entire mineral reserves will be completely exhausted the overburden will be utilized for backfilling in the part of the pit and remaining pit will be allowed to collect seepage and rain water which will act as a temporary reservoir. This activity will be carried out only after obtaining permission from the competent authorities and local community. If the mined out pit is been allowed to behave as a reservoir the same will be done after due procedure for the protection of reservoir.

If permission not obtained for handling of waste, backfilling will be carried out nearly existing ground profile with entire waste material and spread out the preserved topsoil over the backfilled area to facilitate green belt development purpose.

The quarry area will be fenced with barbed wire fencing also safety bund will be formed around the quarry to prevent inadvertent entry of public and cattle (Refer plate No. VII and IX).

4.4.0 METHOD OF MINING

4.4.1 OPEN CAST WORKING

In accordance with the Regulation 106 (2) (b) of the Metalliferous Mines Regulations 1961, in all open cast working where the ore body forms hard rock, the working faces and sides should be adequately benched and sloped; a bench height not exceeding 6m and a bench width not less than the bench height has to be maintained. The slope angle of such benches and sides should not exceed 60° from horizontal. However, observance of these statutory provisions in granite dimensional stone quarrying is difficult to maintain the 60° slope due to the field difficulties and technical reasons as below:

1. Recovery of the granite mineral is to be as undamaged rectangular dimensional blocks. In the attempt to form the benches and sides with the above statutory parameters haphazard blasting may be involved. In which case the commercial granite body may get spoiled inevitably due to generation of blasting cracks.
2. In the exercise of forming the benches with 60° slope within the granite deposit, the portion confined within the 60° as well as its complimentary part in the extricated block will become as mineral waste while shaping them into rectangular blocks.
3. The granite industry need blocks as huge as a few cubic meter volumes with measurements up to 3m x 2m x 2m. Production of such huge blocks in turn increases the recovery and reduces the mineral waste during dressing. Blocks of smaller size of certain varieties of granite are now marketable and have a good commercial value.
4. Formation of too many benches with more height and the width equal to the height may lead to mineral lock up.

Hence, in order to avoid granite waste and to facilitate economical mining operations, it is proposed to obtain relaxation to the provisions of Regulation 106 (2) (b) upto a bench parameter of 5m height & 5m width with vertical faces. Such a provision of relaxation of the Regulation has been provided within the regulation 106 (2) (b). The entire terrain is made up of hard rock, compact sheet and possess high stability on slope even at higher vertical angles. The same will be inspected by the Director of Mines Safety and permission to be obtained under Regulation 106(2)(b) of the Metalliferous Mines Regulation, 1961. The factor of safety is anticipated as 1.2 to 1.5 which has a high stability even if it attains the ultimate pit limit. The overall pit slope angle as proposed will always be maintained from 40° to 50° in order to have a safe movement of men and machinaries working in the quarry. Hence, the slope stability of the quarry is well within the prescribed standards. Gradient of the ramps is proposed in such a way that it is not steeper than 1 in 10 and its length shall be more than 50m at one stretch.

4.4.2 EXTENT OF MECHANIZATION

The following machineries are proposed for the development and production work at this quarry. The proposed and adopted machinery make may variable, if machinery making companies provide upgraded machines the same will be apllied as per availability of market.

Table – 7

I. DRILLING AND CUTTING MACHINE

S.No.	Type	No.	Dia Hole mm	Size Capacity	Make (same or any other similar model)	Motive power
1	Jack hammer	4	35	1.2m to 6m	Atlas Copco	Compressed air
2	Compressor	1	-	400psi	Atlas Copco	Diesel drive
3	Diamond Wire saw	1	-	20m^3/day	Optima	Diesel Generator
4	Diesel Generator	1	-	125kva	Kirloskar	Diesel

Table – 8

II. LOADING EQUIPMENT

S.No.	Type	Nos	Capacity	Make (same or any other similar model)	Motive Power
1	Crawler Crane	1	855	Tata	Diesel Drive
2	Excavator	1	280 HP	Kobelco-350	Diesel Drive

III. HAULAGE WITHIN THE QUARRY & TRANSPORT EQUIPMENT

Table – 9

S.No.	Type	Nos	Capacity	Make	Motive Power
1	Tipper	2	20 tons	Tata	Diesel Drive
2	Water Sprinkler	1	6000 ltrs	Tata	Diesel Drive

IV. TRANSPORT FROM THE QUARRY HEAD TO DESTINATION

Transportation from quarry head to customer destination is done by truck or by trailors.

V. MISCELLANEOUS:

Apart from the above the following tools and tackles are required for quarry operation.

For operation

The operation of granite quarry requires the following loose tools material and have to be kept sufficiently in stock for non - interruption of the quarry work.

1. Drill rods - 0.3m, 0.5m, 0.75m, 1.65m, 2.25m, 3m, 3.6m, 5m upto 9m.
2. Steel Alloy chains of sufficient length of 10m, 12m, 16m, 18m etc., sizes.
3.'D' shackles to link the chain lengths.
4. Rubber hose of required length.
5. Hose clamps to link the compressor delivery hoses.
6. Feather and wedges of 6" and 12" dia sizes utilize for splitting the block from the mother rock. This is an important tool in the operation of a quarry.
7. Crow bars.
8. Spades.
9. Sludge Hammer.
10. Iron Pans.
11. Pitcher Hammer.
12. Chisels.
13. Consumables, such as diesel, Hydraulic oil, grease, abrasive wheels, welding machines etc.,

14. Stock of essential spare parts of machinery.
15. Explosive as per the licensed quantity.
16. Diamond Wire required length.
17. Besides diamond wire saw equipment and new innovative machine specifically for granite with accessories are required to liberate the rock from to parent body to minimize damage and to obtain good recovery.

Splitting the sheet rock by Diamond wire sawing which increases substantial recovery potential. Hence it is proposed to adopt "Diamond wire saw cutting" for best recovery.

The above machineries are adequate to meet out the development and production schedule drawn out during this mining plan period.

5.0 BLASTING:

a. Broad Blasting Parameters:

In general for granite quarrying does not involved deep hole drilling and heavy blasting, only secondary blasting is proposed with Jackhammer drilling by 35mm dia hole for removal of overburden and secondary fragmentation for handling the wastes.

The granite industry needs blocks for about 3m x 2m x 2m for International buyers hence small blocks blasting pattern is not followed. In granite quarrying it is very difficult to prescribe the blasting pattern and charge/ hole as it depends upon the various factors like type of rock, texture, planes of weakness, required size of block, etc. Hence, it is difficult to pronounce a definite pattern of holes with regard to spacing, burden and depth for secondary fragmentation. Hence, only blasting is proposed for secondary fragmentation for handling the wastes and not for production of granite blocks.

Now-a-days Diamond wire saws are used for splitting the blocks from parent sheet mass. It is a new innovative Eco-friendly splitting technique without involving blasting. This is increase the recovery percentage of granite blocks and reduces from induce fissures due to blasting. Besides those are noise free Eco friendly machineries and it optimizes the usage of explosives. Splitting within the sheet rock is by diamond wire sawing which increases substantial recovery potential. Hence it is proposed to follow diamond wire saw cutting for better recovery of granite dimensional stone.

b. Type and use of explosives

In granite quarries, only diamond wiresaw machines are deployed for liberating the granite blocks from parent sheet mass. The waste thus generated after diamond wiresaw cutting will be fragmented by using low intensity explosives. This explosives are used only for size reduction to handle the waste from pit mouth to waste dumps. The aim is to recovery as large block as possible. During future development of quarrying, removal of over burden will be done by blasting with explosives that will be used are NONEL and 25mm Slurry explosives are proposed in 35mm dia holes drilled by Jackhammer.

Hence only low intense explosives like NONEL and Gelatin Sticks are used. Nonel initiation provides reasonably good solution to fly rock problem. The main objectives of Nonel Blasting are to reduce the ground vibration, noise, flyrocks generated due to blasting operations. The overall cost of blasting in NONEL is very less compared to electrical blasting and hence it optimizes the cost of blasting.

c) Storage of explosives:

Authorized explosive dealers supply the explosive at site as per the day's requirement. However, portable magazine 'M' type has been proposed to install in the ear marked places and the applicant is advised to get necessary license from Controller of Explosives (Petroleum and Explosives safety Organization) for storing explosives in the above area after the grant of quarry lease.

The blasting and quarrying operation is advised to handle by competent qualified statutory personnels approved by the DGMS and competent authorities. The Competent Qualified Statutory personnel appointed by the applicant will maintain the records of Explosives as per the Indian Explosives Act, Mines Act and Environment Protection Act.

6.0 MINE DRAINAGE:

a) Depth of water table based on observations from nearby wells and water bodies:

Water level is found at a depth of 131m - 136m, which is observed with water level indicator from the nearby bore wells. The water table fluctuation is observing the water levels during rainy seasons in the nearby wells.

b) Depth of Working:

The quarry operations are confined to well above the water table. The quarry operation is proposed to a depth of 26m – below from the ground level.

c) Quantity and Quality of Water Likely to be encountered, the pumping arrangements and places where the mine water is finally proposed to be discharged:

The multi-colour granite consisting of no chemical effluent and toxic elements, hence there will be effluent discharge from Granite quarry is non-toxic. Hence the quality of water would not be affected by mining activities. Anyhow, Garland drain will be constructed all along the boundary to prevent surface run-off water entering into the quarry pit. If seepage water may encounter during rainy season, the same will be pumped out from the pit to settling tank through the garland drain by 5 HP motor pumps. After settling of particles from the drained water, the same will be utilized for Greenbelt development and water sprinkling on dust prone areas through the water sprinkler.

7.0 STACKING OF MINERAL REJECTS AND DISPOSAL OF WASTE:

i) Nature of Quality of Topsoil:

The topsoil is reddish gravely in nature. It occurs to a depth of 2m(avg). About 17,952 MT of topsoil would be generated during this plan period. The same will be preserved along the safety zone and in the non-mining area and it will be utilized for construction of bund and greenbelt development.

ii) Nature of Over Burden:

Topsoil and Weathered rock is the overburden in the applied area.

iii) Nature of Waste/ Rejects generate during first five years:

The Mineral Rejects and Weathered rock is the mine waste. The Proposed generation of waste for first five years are given table below.

Table – 10

Years	Mineral Rejects @ 40% (MT)	Weathered rock (MT)	**Total Waste (MT)**	Topsoil (MT)
I	10,048.8	12,480	**22,528.8**	8,184
II	10,350.9	8,160	**18,510.9**	4,488
III	11,090.0	9,600	**20,690.0**	5,280
IV	9,858.0	-	**9,858.0**	-
V	8,856.3	-	**8,856.3**	-
Total	**50,204.0**	**30,240**	**80,444.0**	**17,952**

iv) Sub Grade Mineral:

There is no sub grade mineral.

v) Land Chosen for Disposal of Waste / Manner of Disposal of Waste / Year wise:

During the mining plan period the quarried out waste will be proposed to dump on the Northern side.

The maximum dimension of the proposed waste dump is given table below (Refer Plate Nos. V, VI & VII).

Table – 11

	Area(m^2)	Height(m)
Proposed Waste Dump	1,950	17.46

Manner of Disposal of Waste:

As and when there is accumulation of waste, the same is loaded into the tipper by loading machines and dumped in the respective places ear-marked for the purpose.

The waste management plan with reference to the quantum of waste generated is shown in quarry layout and Afforestation plan (Plate No.VI).

There is no slurry anticipated in the quarry operation. Besides the granite waste does not produce any toxic effluent in the form of solid, liquid or gas.

8.0 USE OF THE MINERAL:

a) Describe briefly the End Use of the Mineral:

The quarried out Grey Granite blocks are either exported as rough blocks or processed in designated factories as value added products such as slabs, tiles, fancy items and, precision surface plates for construction and engineering application.

b) Indicate physical and chemical specifications stipulated by buyers:

No chemical parameters specified by the buyers, but the buyers are preferred Gang saw size blocks of dimensional stone such as having measurements of 3m x 2m x 2m.

i. Quality Control:

The Grey Granite deposit occurring in this quarry shows uniform quality throughout and hence quarried and marketed as a single variety.

The exploited blocks are carefully examined for any natural defects such as joints, cracks, xenoliths, secondary Pegmatitic growth etc and such defects is removed manually using feather and wedges and the blocks are then shaped into perfect rectangular dimensional stone blocks by chiseling. Different price for each quality material have been fixed and the entire production quantity is marketed accordingly.

ii. Sale Price of Mineral:

The sale prize of the grey granite dimensional stones varies from Rs. 5,000 to Rs. 10,000/ MT based on their respective sizes at present market scenario.

iii. Captive Consumption:

No captive use, the quarried out Grey Granite blocks will be sold to the needy customers through out the contry in the form of required dimentional blocks. No processing is proposed in the applied area.

iv. Export or Industrial use:

The Granite dimensional stone is proposed to either exported as rough blocks or trading as raw form to the needy domestic customers for making value added products.

v. Surface Transport

The mode of transport of the granite blocks produced and marketed is by road to various customer destinations and granite processing units located at different parts of the country. The Grey Granite blocks approved for export market are shipped from Chennai Port to various countries and if required the blocks may be shifted from Thoothukudi Port which depend upon the exporter's destination from time to time.

9.0 OTHERS:

9.1 Site Services:

The simple methods adopted and the limited scale of activities involved in granite dimensional stone quarrying does not require High Tension Electric Power supply or huge workshop facilities. The quarry office, storeroom, first aid room, Rest room, Hygienic modern Sanitary etc. are proposed to be constructed as semi-permanent structures within the lease applied area (Please refer Plate No. VI). The quarrying operation is restricted to one general shift during day time only. Major Machinery repair works are attended at Krishnagiri (8km-NW) and minor repairs are carried out by the company's personnel at the quarry site itself.

9.1.1 Welfare measures:

a) Drinking water:

Packaged Drinking water will be brought from the water venders in the Krishnagiri which is located at 8km on the Northwestern side of the area.

b) First Aid facility:

First aid kits are kept in Mines office room, in case of such eventuality is the victim will be given first aid immediately at the site by the competent and statutory foreman/permit manager will be in charge of first aid and injured person will be taken to the hospital by the company's vehicle.

c) Labors Health:

Periodically medical check-up related to occupational health safety will be conducted to all the workers in company's own cost.

d) Precautionary safety measures to the Workers:

All Personal Protective Device with Safety Helmets, Mine Goggles, Ear Plugs, Ear Muffs, Nose Mask, Reflector Jackets and Safety Shoes will be provided to the workers as per the specification of Director of Mine Safety. Besides also conducting pre medical checkup for all workers for any health related problems proper training and induction will be given to quarrying personal to create awareness of conventional Eco friendly quarrying operations. Also the applicant ensure to registered in ongoing Labour Welfare Board insurance scheme to all the workers working under its control.



- Helmets,
- Reflector Jackets,
- Dust mask,
- Mine Goggles,
- Ear plugs,
- Ear muffs,
- Safety Shoes.

Table – 12

<table>
<tr><th>S. No.</th><th>Description- Nearest</th><th>Particulars</th><th>Aerial Distance & Direction</th></tr>
<tr><td>1</td><td>National Highway</td><td>Krishnagiri – Uthangarai (NH – 77)</td><td>400m – North</td></tr>
<tr><td>2</td><td>State Highways</td><td>Krishnagiri – Tirupathur (SH – 131)</td><td>8.2 km – NE</td></tr>
<tr><td>3</td><td>Railway station</td><td>Samalpatti</td><td>28.5 km – SE</td></tr>
<tr><td>4</td><td>Airport</td><td>Bengaluru</td><td>100 km – NW</td></tr>
<tr><td>5</td><td>Habitation</td><td colspan="2">350m – SW</td></tr>
<tr><td>6</td><td>Town</td><td>Krishnagiri</td><td>8 km – NW</td></tr>
<tr><td>7</td><td>Government School</td><td>Chendarapalli – P.U.P. School</td><td>830m – East</td></tr>
<tr><td>8</td><td>Government Hospital</td><td>Krishnagiri</td><td>8 km – NW</td></tr>
<tr><td>9</td><td>Reserved Forest</td><td colspan="2">Thogarapalli R.F. 4.15km – SE. (Refer Annexure No.X).</td></tr>
<tr><td>10</td><td>Defense Installation/Historical Monuments/ Archaeological</td><td colspan="2">Nil within 500m radius.</td></tr>
<tr><td>11</td><td>Water Bodies</td><td colspan="2">Odai – 50m – Eastern side.
Tank – 60m – Western side.</td></tr>
<tr><td>12</td><td>Interstate Boundary</td><td colspan="2">Around 17.9 km – NE (Andhra pradesh border).</td></tr>
<tr><td>13</td><td>Critically Polluted areas identified by the CPCB</td><td colspan="2">Around 74 km – NW (Bengaluru, Jigani – KIADB).</td></tr>
<tr><td>14</td><td>Protected areas Notified under wildlife (Protection) Act,1972</td><td colspan="2">Around 34.8 km – West (Cauvery North Wildlife sanctuary) Eco Sensitive zone – 33.8 km – West.</td></tr>
<tr><td>15</td><td>Applicability of CRZ, Notification 2011 as amended.</td><td colspan="2">Not Applicable.</td></tr>
<tr><td>16</td><td>Applicability of Hill Area Conservation Authority (HACA) Clearance.</td><td colspan="2">Not Applicable.</td></tr>
<tr><td>17</td><td>Housing area, EB line (HT & LT Line)</td><td colspan="2">There is no EB-LT/HT line or Housing area situated within 50m radius.</td></tr>
<tr><td>18</td><td>Boundaries of the permitted area.</td><td colspan="2">The boundaries of the permitted areas are as follows (Refer Plate No. II):
North - S.F.No. 223/1A1 (P)
East - S.F.Nos. 223/1B1,223/1B2, 223/1A2 & 224/2B1.
South - S.F.Nos. 224/2A (P) & 228.
West - S.F.Nos. 228 & 223/1A1(P).</td></tr>
<tr><td>19</td><td>Adjacent Patta lands / Govt. Land</td><td colspan="2"><table><tr><th>Direction</th><th>Classification</th><th>Safety Distance</th></tr><tr><td>North</td><td>Patta</td><td>7.5m</td></tr><tr><td>East</td><td>Patta</td><td>7.5m</td></tr><tr><td>South</td><td>Patta</td><td>7.5m</td></tr><tr><td>West</td><td>Patta</td><td>7.5m</td></tr></table>(Refer Plate No. II).</td></tr>
</table>

9.2 Employment Potential:

The following manpower for machinaries as well as for operational activities are proposed to carry out the day-to-day quarrying activities at the peak production capacity aimed and also to comply with the statutory provisions of the metalliferous mines regulations, 1961.

Table – 13

Designation	Present Employment position	Employees Requirement	Total
a) Supervisory category			
Mine Manager	-	1	1
Mine Foreman	-	1	1
Geologist	-	1	1
Blaster/Mate	-	1	1
b) Skilled labour			
Excavator Operator	-	1	1
Crane Operator	-	1	1
Wire Saw Operator		1	1
Tipper Driver	-	2	2
Water sprinkler Driver	-	1	1
Jack-Hammer Drillers	-	12	12
c) Unskilled			
Security	-	1	1
Helper	-	2	2
Co-operator and Cleaner	-	5	5
Total	-	**30**	**30**

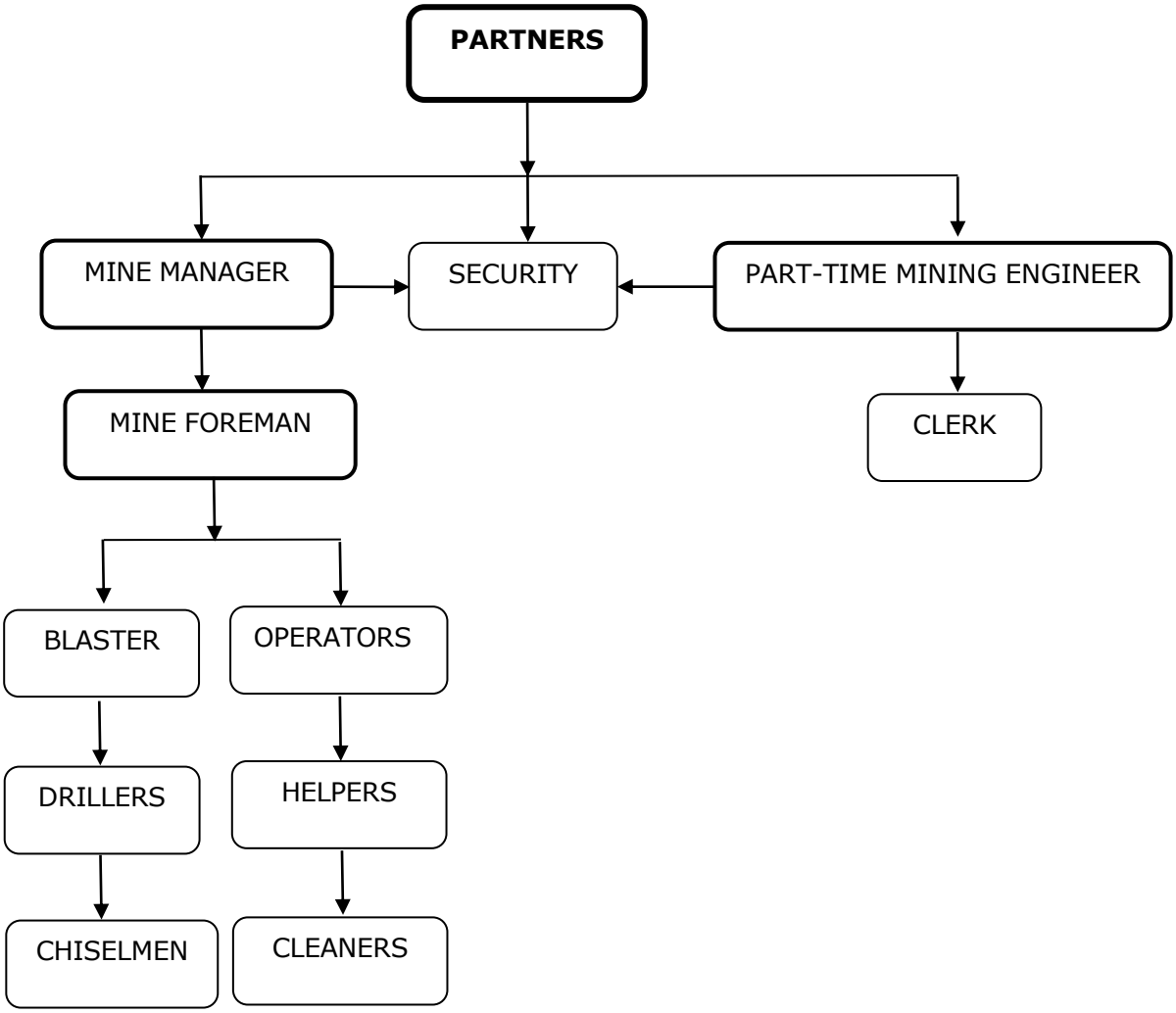
The above manpower is adequate to meet out the production schedule and the machinery strength envisaged in the mining plan and also to comply with the statutory provisions of the Mines Safety Regulations.

The proposed output per man shift:

Table – 14

Annual Peak Production @ 40%		7,394 MT
No. of days likely to be worked		300 days
Average production per day		24.6 MT
OMS =	Daily Production/Daily employment	24.6/30 = 0.821 MT

ORGANIZATION CHART



10.0 MINERAL PROCESSING:

No mineral processing/beneficiation involved.

The exploited blocks are carefully examined for any natural defects such as joints, cracks, xenoliths, secondary Pegmatitic growth etc and such defects is removed manually using feather and wedges and the blocks are then shaped into perfect rectangular dimensional stone blocks by manual chiseling.

11.0 ENVIRONMENTAL MANAGEMENT PLAN:

11.1 Baseline Information:

The EMP is prepared based on the Mines act, Rules & amendment from by state & central government. If the SEIAA/SEAC instructed the modification and alter the EMP the outcome of their recon would be final and the applicant is instructed to followed the EIA / EMP for its compliance as per the CPCB / TNPCB Norms.

i) Existing Land Use Patterns:

It is a barren land, the area was not utilized any purpose.

Table – 15

Description	Area at present (Ha)	Utilized (%)
Area under quarry	Nil	-
Waste dump	Nil	-
Infrastructure	Nil	-
Roads	Nil	-
Green Belt	Nil	-
Unutilized	1.80.0	100
Grand Total	**1.80.0**	**100**

ii) Water regime:

Ground water is about 57m depth below ground level. The quarrying is restricted up to 26m-BGL, hence the quarry operation will not intersect the ground water. Garland drains will be constructed all along the boundary barriers to prevent the surface water flow inside the quarry and also to divert the surface water flow to the nearby natural drains.

iii) Flora and Fauna:

The main floras are observed in the area is Brinjal, Cassava, groundnut, Paddy, Palm, Coconut, Mango, Neem, Castor, Calotropis and shrubs are observed around the area. No plants of botanical interest or scheduled fauna are recorded within the impact zone.

iv) Climatic conditions:

The area receives an average rainfall of about 851mm/per annum and the rainy season is mainly from Oct - Jan during North East, monsoon. The summer is hot with maximum temperature of 42°C and winter encounters a minimum temperature of 24°C.

v) Human settlement:

There is no approved habitation located within 300m radius of the area. There are few villages located within 5km radius of the lease area. The direction, approximate distance and population are given below.

Table – 16

S.No.	Name of the Village	Direction	Approximate Distance	Approximate population
1.	Chendarapalli	NE	1km	6,500
2.	Soolamalai	NW	1km	2,000
3.	Modikuppam	South	2km	2,550
4.	Orappam	NW	4km	6,800

vi) Public buildings, Monuments and Places of Worships:

There is no public building, place of worship or archaeological or Historical monument within 300m radius of the area.

vii) Weather the area falls under notified area under water (Prevention and Control of Pollution) act, 1974:

The area falls under notified area under water (Prevention and Control of Pollution) Act, 1974.

11.2 Environment Impact Assessment:

The mining plan is proposed for very small production of granite dimensional stone without involving deep hole drilling and heavy blasting. Such limited quarrying activity is not likely to cause any significant impact adversely on environment as far as pollution of air, water and noise is concerned. If any significant variation found during the Half year compliance, the Environment Management Plan will be altered accordingly. Anyhow, the impacts are anticipated due to the mining operations and its mitigation measures are given below:

Environment	Anticipated impact	Mitigation measure
Land Environment	i. Topography of the area will continually change till the entire life of quarry.	i. Backfilling is only mitigation measure for topographical changes. The part of the mine pit will be backfilled with mined waste at the end of life of quarry and preserved topsoil will be spread out over the backfilled area for reclamation and rehabilitation. If permission is granted by the state Government for disposal of waste to the crushing unit, there is no possibility of Backfilling. The mined out pits will act as a water reservoir. ii. The excavated benches shall be developed for plantation with grasses, herbs and shrubs of local species to improve aesthetic of the area and to prevent any soil erosion and landslide. iii. Mining benches will not exceed beyond the approved height and width.
	ii. Quality of soil on surrounding land environment will impact due to: a) Dust propagation during quarry operation like drilling, blasting, loading, unloading and movement of men, Vehicle and machineries. b) Vehicular and machineries emissions. c) Non-Degradable waste materials dumping like polythene bag, jute bag, Plastic water and cool drink containers. iii. Inappropriate dumping of mined out soil and waste will impact the Soil quality of non-mining area.	i. Quarried out topsoil will be preserved separately in non-mining area (safety barriers) also grass cover and plantation will be forming to sustain the quality and quantity of soil also soil based organisms. ii. Regular water sprinkling on dust prone area like haul road and other active area to arrest dust generation. iii. Green belt will be developed in the safety barriers with thick long leave plants to prevent dust and vehicular emissions propogation. iv. Wet drilling with dust extractor unit will be carried out and quarry material will be handled under wet condition to minimize the dust generation. v. Non-Degradable materials like polythene bag, useless articles, plastic water and cool drink containers will not be allowed within quarry premises. vi. The waste material will be properly dumped in area earmarked for the purpose only to sustain the quality of soil on non-mining area.
Water Environment	i. Alteration of natural flow direction of surface run off rain water during rainy season due to modification of topography.	i. Garland drain with check dam will be constructed around the quarry to prevent surface run off rain water entering into the pit. ii. The waste material will be properly dumped on area earmarked for the purpose

	ii. Inappropriate dumping of waste material shall affect the natual flow direction of surface run off rain water. iii. Contamination of surface water course due to dust and vehicle emission propagation due to quarry operation and movement of machinery and Vehicles. iv. A small amount of water will be contaminated due to utilization of water for wire saw cutting.	only and retaining wall with weep holes will be constructed around the dump. iii. Mining will be completely stopped during the monsoon for free flow of surface run off and allowing natural recharge of groundwater. iv. The pit water collected due to rain will be utilized for water sprinkling on the haul Roads and plantations after settling of silt and other particles. v. Septic tanks and soak pits will be provided for the disposal of domestic/ washroom effluents. The sewage water will be collected from the soak pit and disposed to the authorized wastewater treatment plant. vi. The contaminated water during wire saw cutting will be collected in a sump and the same will be recycled to reduce the water consumption.
Air Environment	In surface mining operations, the source of air pollution may cause deterioration of air quality due to the fugitive dust emissions from drilling, blasting, scooping, loading-unloading operation of extracted mineral and its transportation. Drilling/blasting and loading of quarry material would be associated with the fugitive dust emission in the active area whereas fugitive emission during transportation would affect the areas/villages situated adjacent to the road side. Another source of air pollution would be emission from the drilling machinery and excavators/Trucks/vehicles to be used for loading.	i. Regular water sprinkling to the haul road to arrest the dust generation. ii. Green belt will be developed in the safety barrier which will be act as an acoustic barrier, the plants with thick long leaves to arrest the fugitive dust and vehicular emissions propagations. iii. Diamond wire saw cutting method does not produce dust. Anyhow, during development wet drilling/ dust extractor unit will be carried out to minimize the dust generation. iv. Controlled blasting with Nonel initiation by Proper blasting techniques will be followed for removing the waste during development to reduce the fine dust propagation. v. Quarry material will be handled under wet condition during loading and unloading to minimize the dust generation, besides loaded materials are covered by Tarpaulin until reaches its destination during transportation. vi. Provision of dust filters/ mask (PPE) to workers working at dust prone areas. vii. Vehicular emission as a result of combustion of diesel generates small particulate matter (PM_{10} & $PM_{2.5}$), Nitrogen oxides and Sulphur dioxide (NO_2 & SO_2). High quality diesel will be used in the motor vehicles to control these pollutants.

		viii. PUC (Pollution under control) certified vehicles will be used for transportation. ix. CPCB Prescribed emission standards for the vehicles would be followed. x. All vehicles and their exhausts would be well maintained and regularly tested for pollutant concentrations.
Noise and Vibration	In the present mining activity for building material, noise will be generated from drilling machinery, blasting and vehicular movement. Noise level in the working environment is compared with the standards prescribed by Central Pollution Control Board as adopted and enforced by the Govt. of India through Noise Pollution (Regulation and Control) Rules, 2000.	i. Selection of new low – noise equipments under BS-VI for the quarry operation. ii. The noise levels shall be maintained within the permissible levels by involving all the noise regulating measures in vehicles and drilling/blasting operations. iii. Diamond wire saw cutting is proposed for mining activity, which is eco-friendly noise and vibration free machine. Anyhow, to ensure minimum noise and vibrations due to blasting during mine development, NONEL delay blasting is proposed to control the Noise, Vibration and fly rock. iv. Personnel Protective Equipment (PPE) like earmuffs and earplugs shall be provided to the employees whose in critical operation like drilling, blasting and excavation as occupational safety measures. v. Proper maintenance done with regular interval by the Oiling and greasing for the machineries and vehicles to control the Source of noise during operation and transportation. vi. Transporting vehicles are enforcing the speed limits of 25km/hour within quarry area and not exceed 40km per hour from quarry to destination for reduce Noise and vibration level. vii. There would be restrictions on mining activity and vehicular movement during night hours.
Biological Environment	No plants of botanical interest or animals of zoological interest recorded within 500m radius. The anticipated impacts on biological environment as follows: i. Public and cattle may cause injury by inadvertent entry to the pit. ii. Disrupt the living insects in the overburden due to mining	i. The barbed wire fencing and safety bund will be constructed around the quarry to prevent inadvertent entry of public and cattle. ii. The natural habitats of the existing flora and fauna will not be disturbed by adopting well maintained machineries, equipments and diamond wire cutting for mining. iii. Fruit bearing trees will be planted to survive the existing native Faunas.

	activity also which will induce the diversity of existing ecosystem. iii. Natural habitats of the existing faunas will change due to the noise and vibration during operation. iv. Deposition of dust on the plant and crop leaves is affecting the photosynthesis, Pollination, growth ratio and agriculture productivity.	iv. There is no proposal for cutting any trees within the proposed area. v. Water sprinkling on haul roads would be arrest the dust generation, thus avoiding damage to the crops and plants.
Socio – Economic Environment	Any activity during mining will have adverse impact on Environment, careful mitigation measures are proposed to balance the impact on the existing environment and the applicant is always instruct to carry out safe, sustainable, eco-friendly mining operations at all times. The following positive impact on the society due to this mining activity. i. It is proposed to provide employment to about 30 persons for carrying out mining operations and give preference for employment to the local peoples. In addition, there will be opportunity for indirect employment to many people in the form of contractual jobs, business opportunities, service facilities etc. ii. The employment opportunities both direct and indirect will contribute to enhanced money incomes to job seekers with minimal skill sets especially among the local communities and their economic status will be enhanced. iii. Improvement of Physical structure like Road Transport facilities, Communications, Medical, Educational and social benefits will be made available to the nearby civilian in addition to the workmen employed in the mine. iv. The continuation of opportunity for the employments, the nearby villages, living peoples and their life style would be improved. v. The applicant is advised to invest the CER cost (@ 2% from the total Project Cost) to develop the local Panchayat.	i. Preference employement will be given to the local community. This quarring activity will also indirectly provide employement to the local community by the way of petty shops, hotels, etc. ii. The CER activity proposed by the applicant will enhance the requirement of the local community. iii. The improvement of physical structure like road transport facilities, medical, social benefits will get enhanced. iv. There will be a positive enhancement of lifestyle of the nearby village.

11.3 Monitoring schedule for different Environmental Components after commencement of mining and other activities:

i) Environmental impact assessment statement describing impact of mining on the next five years:

The mining plan is proposed for small production of granite dimensional stone without involving deep hole drilling and heavy blasting. Such limited quarrying activity is not likely to cause any impact adversely on environment. As far as pollution of air, water and noise concerned, the environmental impact studies will be conducted as per EIA notification issued by MoEF& CC. It is B Category mine. The compliance monitoring will be carried out for every six months as prescribed by the MOEF&CC and with state concerned authorities.

ii) Proposal for waste management:

The mine waste in the mine includes, rock fragments, rock chips, rubbles generated as mineral waste during production work.

The waste produced during this mining plan period (80,444 MT) will be proposed to dump on the Northern side with maximum dimension of (Area) 1,950m x (Height) 17.46m. The quarried out topsoil will be preserved all along the safety barrier and utilized for formation of bund and green belt purpose.

The waste management plan with reference to the quantum of waste generated is shown in quarry layout plan (Please refer Plate No.VI). The maintenance of machineries & fueling will be carried out as per the TNPCB Norms and the waste will be disposed in manner.

iii) Proposal for reclamation of land affected during mining activities and at the end of mining (refilling / fencing etc.):

In the proposed mining plan only 26m depth has been envisaged as workable depth for safe & economic quarrying for the entire lease period. After expiry of the lease period, if the mineral reserves available and Market persist, the applicant may apply a renewal of quarry lease as to develop and conserve mineral reserves. If permission is granted for removal of waste, the waste material will be supplied to needy crusher for convert to the M-Sand, building and road construction material from concerned authorities after paying the seniorage fee. When the entire mineral reserves will be completely exhausted if permission obtained for disposal of waste the remaining overburden will be utilized for backfilling in the part of the pit and remaining pit will be allowed to collect seepage and rain water which will act as a temporary reservoir. If permission not obtained for handling of waste, backfilling will be carried out nearly existing ground profile with entire waste material and spread out the preserved topsoil over the backfilled area to facilitate green belt development purpose.

The quarry area will be fenced with barbed wire fencing also safety bund formed around the quarry to prevent inadvertent entry of public and cattle (Refer plate No. VII and IX).

iv) Programme of Greenbelt development (indicate extend, number, name of species to be afforested):

The safety distance all along the boundary barrier will be utilize for subsequent greenbelt development. It is proposed to plant 500 tree saplings around 2,810m^2 inside the quarry lease area and 500 tree saplings in the haul road during the first two years of this plan period and maintain at least 800 plants at the end of the lease period at a survival rate of 80%. As per the SEIAA recommendation, the plantation will be carried out based on the output Environmental Clearance and the recommended species will be carried out for green belt development. Appropriate species of trees will be planted in a phased manner as described below.

Table – 17

Year	No. of tress proposed to be planted		Area to be covered in applied area (m^2)	Name of the species
	Safety zone	Approach Road		
I	100	250	562	Poovarasu, Sarakondrai, Mookuchali maram, Mamaram, Thespesia populnea, Nuna, Mantharai, etc.,
II	100	250	562	
III	100	-	562	
IV	100	-	562	
V	100	-	562	

11.4 Proposed financial estimate / budget:

Proposed Financial Estimate for Quarry operations and Environment Management (EMP).

Table – 18

A. Machineries and Equipment Cost

S.No.	Description	Approximate Cost (Rs.)
1.	Land Cost (As per Govt. Guideline value at present) S.F.No. / Extent (Ha.) / Cost/Ha.(Rs.) / Cost for area (Rs.) 223/1A1(P) / 1.38.0 / 12,38,000 / 17,08,440 224/2A1 / 0.42.0 / 12,38,000 / 5,19,960 **Total** / **1.80.0** / / **22,28,400** Round off = Rs. 22,29,000/- (source : https://tnreginet.gov.in/portal/)	22,29,000
2.	Labour Shed	5,00,000
3.	Sanitary Facility	1,00,000
4.	Excavator (1 No.)	50,00,000
5.	Crawler Crane (1 No.)	50,00,000
6.	Tipper (2 Nos.)	50,00,000
7.	Wire Saw (1 No.)	4,00,000
8.	Diesel Generator (1 No.)	7,50,000
9.	Compressor with loose tools (1 No.)	4,00,000
10.	Jack Hammer (4 Nos.)	2,00,000
11.	Drinking Water Facility	1,13,000
Total Machineries and Equipment Cost		**1,96,92,000**

B. Proposed financial estimate / budget for (EMP) Environmental Management Plan:

Budget Provision for the first five years period

B. EMP Cost:		**Cost (Rs.)**
1	Environmental Monitoring Program (Air Quality, Water Quality Sampling, Noise and Ground vibration test) Rs. 50,000/ year.	2,50,000

2	Water will be sprinkled in the haul roads by own water sprinkler for dust suppression. The cost would be around.		6,00,000
3	Fencing will be construct to prevent inadvertent entry of public and cattle, the cost is (650m length x Rs. 300/- per meter).		1,95,000
4	Construction of Garland drain with check dam to prevent surface run-off rain water entering to the quarry pit, the construction cost is around (Total Peripheral length 590m x Rs. 300/meter).		1,77,000
5	First aid room & accessories		50,000
6	Dust extractor unit for Jackhammer (4 Nos.).		1,00,000
7	Safety Kits for workers (Personnel Protective Equipment).		50,000
8	Greenbelt development for dust suppression & Noise control measures	Greenbelt development will be carried out along the safety barrier; the cost would be around (500 saplings x Rs. 200/sapling)	1,00,000
		Greenbelt development will be carried out on haul road; the cost would be around (500 saplings x Rs. 200/sapling).	1,00,000
Total EMP Cost			**16,22,000**

The EMP cost for the five years plan period would be around **Rs. 16,22,000/-**

Total Cost of the Project including EMP Cost	
Description	**Cost (Rs.)**
A. Machineries and Equipment Cost	**1,96,92,000**
B. EMP Cost	**16,22,000**
Total Project Cost (A+B)	**2,13,14,000**
C. The company will ensure to carry out the Corporate Environment Responsibilities **(CER)** as recommended by the concerned authority.	

(Total project cost including EMP cost is about rupees two crores thirteen lakh and fourteen thousand only).

12.0 PROGRESSIVE QUARRY CLOSURE PLAN:

12.1 Introduction:

The Progressive Quarry Closure Plan for Grey Granite quarry lease applied area over an extent 1.80.0 Hectares of Patta lands in S.F.Nos. 223/1A1 (Part) and 224/2A1 (Part) of Chendarapalli Village, Bargur Taluk, Krishnagiri District, Tamil Nadu State has been prepared for **M/s. AB Granite,** having an office at No.223/A, Chendarapalli Village, Anchoor Post, Bargur Taluk, Krishnagiri District, Tamil Nadu – 635 203.

12.2 Present Land use pattern:

Land Use Table – 19

Description	Present area in (Ha)
Area under Quarry	Nil
Dump	Nil
Infrastructure	Nil
Roads	Nil
Green Belt	Nil
Unutilized	1.80.0
Grand Total	**1.80.0**

12.3 Reasons for closure:

The mineral is not going to be exhausted during the proposed Mining Plan period hence, immediate closure is not planned due to sufficient reserves are available for the entire life of quarry. Hence, the reason for closure will be discussed in Final Mine Closure Plan.

12.4 Statutory obligations:

The applicant ensures to comply all the conditions stipulated in the precise area communication letter before grant of quarry lease and during the course of quarry operations as per the DGMS, Department of Geology and Mines, Labour Enforcement officer, controller of Explosives etc., circulars, Norms, Rules, Regulations and Act.

12.5 Progressive quarry closure plan preparation:

Name and address of the Qualified Person who prepared the progressive closure plan and executing agency who is involved in the Preparation of progressive quarry closure plan.

Name	:	**B. Vengadagiri, M.Sc.,**
		Qualified Person (Under Rule 15(I)(a) and (I)(b) of MCR, 2016)
Address	:	No.105, 5th Cross, Alagapuram
		Salem District
State with PIN Code	:	Tamil Nadu – 636 010.
Mobile	:	+91 86953 32233
Email	:	vengatb6@gmail.com

The applicant will himself implement the closure plan; no outside agency will be involved.

12.6 Review of Implementation of Mining Plan including Progressive Closure Plan upto the Final Closure Plan:

The Mining Plan and Progressive quarry closure plan are being submitted for the first time. In the mining plan is discussed for Reclamation and Rehabilitation will be carried out only at the end of life of quarry. The Grey granite mineral reserves are available for the entire life of quarry. Hence, review of implementation of progressive quarry closure does not arise at present. However, if any work done for progressive quarry closure during this Mining Plan period, it will be discuss an ensuing Scheme period.

12.7 Closure Plan:

(i) Mined Out Land:

At the end of mining plan period the quarry operation carried out 0.47.13Ha area to a depth of 26m out of 1.33.93 Ha of total mineable area upto a depth of 26m. When the remaining reserves will be completely exhausted, the mine closure plan will be prepared and submitted to the competent authority to obtain approval and the same will be implemented. Land use at various stages is given in the table below.

Land use pattern

Table – 20

Description	Present area (Ha)	Area required during Mining Plan period (ha)	Area at the end of life of quarry (Ha)
Area under quarry	Nil	0.47.13	1.33.93
Waste dump	Nil	0.19.50	**#** Backfilled
Site Services	Nil	0.01.00	0.02.00
Roads	Nil	0.01.00	0.02.00
Green Belt	Nil	0.28.10	0.28.10
Unutilized/ Stocking Blocks	1.80.00	0.83.27	0.13.97
Grand Total	**1.80.00**	**1.80.00**	**1.80.00**

If permission is granted for disposal of waste from the State Government, the quarried out overburden only utilized for backfilling. If permission not obtained for disposal of waste, backfilling will be carried out with total waste and spread out the preserved topsoil to facilitate green belt development in the backfilled area.

(ii) Disposal of mining machinery:

All the Machineries are purchased afresh, the same has been maintained in good condition during entire life of quarry, if required the machinaries will be replaced or repaired. After completion of quarry operation all machineries will be utilized at another quarry area or sold out to the second hand. Hence, disposal or decommissioning of mining machinery does not arise.

(iii) Safety & Security:

Safety measures will be implemented to prevent access in the excavation area an un-authorized persons as per Mine Act 1952, MMR 1961.

- Safety measures will be implemented as per Mine Act 1952, MMR 1961, and Mines Rules 1955.
- Provisions of MMR 1961 shall be strictly followed and all roads shall be wider than the height of the bench or equal to the height of the bench and have a gradient of not more than 1 in 16.
- The bench height will be 5.0m if the applicant intends to increase the height of the bench, necessary permission will be obtained from the concerned authority and implemented.
- Width of working bench will be kept about 5.0m for ease of operations and provide sufficient room for the movement of equipments.
- Personnel Protective equipment like dust masks, ear-plugs/ muffs and other equipments shall be provided to the working personnel.
- Notices giving warning to prevent inadvertent entry of persons shall be displayed at all conspicuous places and in particular near mine entries. Sufficient caution and sign boards will be kept in and around the quarry to induct public for awareness.
- Blasting will be carried out in a specific time after giving sufficient caution to the public such as danger signs shall be displayed near the excavations and siren alarm signal will be provide before small amount of blasting time for precautionary action of accident (blasting is carried out only for secondary fragments and not to liberate the Granite body from the parent rock mass).
- Security guards will be posted to prevent inadvertent entry of public.
- In the event of temporary closer, approaches will be fenced off and notice displayed.
- Installation of CCTV cameras in the quarry and entrance of the quarry.
- Monitoring of Quarrying operation by accredited external agency as directed by authorities.

(iv) Disaster Management and Risk Assessment:

This should deal with action plan for high risk accidents like landslides, subsidence, flood, fire, seismic activities, tailing dam failures etc. and emergency plan proposed for quick evacuation, ameliorative measures to be taken etc. The capability of applicant to meet such eventualities and the assistance to be required from the local authorities should be described.

- The mechanized mining activities in the area may involve any high risk accident due to side falls/collapse.
- The complete mining operation will be carried out under the Management and control of experienced and qualified Mines Manager having Certificate of Competency to manage the mines granted by DGMS.
- All the provisions of Mines Act 1952, MMR 1961 and Mines Rules 1955, TNMMCR 1959 and other laws applicable to mine will be strictly complied with.

- During heavy rainfall the mining activities will be temporarily suspended.
- All persons in supervisory capacity will be provided with proper communication facilities.
- Competent persons will be provided FIRST AID kits which they will always carry.

Environmental Monitoring Cell:

A dedicated team nominated by the mine manager or Agent will monitor and maintaine the environmental compliances of the quarry as per the approved Environment Management Plan and report the Compliance to the competant authority from time to time.

Disaster Management Cell:

The Competent Qualified Statutory managers appointed by the applicant as per the Director of Mines Safety will be responsible for Disaster Management. It care of any eventualities his mobile number will be displayed and he will take all the precautions and safety measures as per Mines and Minerals (Development and Regulations) Act, 1957.

(v) Care and Maintenance during Temporary Discontinuance:

In case of any temporary discontinuance due to court order or due to statutory requirement or any other unforeseen circumstance following measures shall be taken for care, maintenance and monitoring of conditions.

- Notice of temporary discontinuance of work in mine shall be given to the DGMS as per the MMR 1961.
- All the mining machinery shall be shifted to a safe place.
- Entrance to the mine or part of the mine, to be discontinued shall be fenced off. Fencing shall be as per the circular 11/1959 from DGMS.
- Security Guards shall be posted for the safety and to prevent any unauthorized entry to the area.
- Carry out regular maintenance of the facilities/area detailed below in such a way as would have been done as if the mines were operation:

Mine roads and approach roads,
Fencing on approach roads,
Checking and maintenance of machines and equipment,
Drinking water arrangements,
Mine office, first aid stations etc.

- Competent persons shall inspect the area regularly.
- Air, water and other environmental monitoring shall be carried out as per CPCB Guideline.
- Care and upkeep of plantation shall be carried out on regular basis.
- Status of the working and status monitoring for re-opening of the mines shall be discussed daily.

In case of discontinuance due to any natural calamities/abnormal conditions, mining operation will be restarted as early as possible after completing rescue work, restoring safety and security, repairs of roads etc.

(vi) Economic Repercussion of Closure of Quarry and manpower Retrenchments:

The quarry lease is granted for a period of twenty years only. As per the production Program envisaged, there will be no effect on the man power as the majority of persons belong to nearby villages and will have an option either to be available for employment for the next contract/ lease or do the agriculture in their fields.

(vii) Time Scheduling For Abandonment:

The lease applied area has enormous potential for continuance of operations even after expiry of the lease period. The details of time schedule of all abandonment will be given at the time of final closure plan.

(viii) Abandonment Cost:

As at present mining is not going to be closed, so abandonment cost could not be assessed. However, based on the progressive quarry closure activities during the plan period, cost is assessed as given below:

Table – 21

ACTIVITY		YEAR					RATE	Cost (Rs.)
		I	II	III	IV	V		
Plantation proposed under safety zone	Nos.	60	60	60	60	60	Rs.200 Per sapling	60,000
	Cost	12,000	12,000	12,000	12,000	12,000		
Plantation proposed in the approach road	Nos.	100	100	-	-	-		40,000
	Cost	20,000	20,000	-	-	-		
Barbed wire fencing 600 Mtrs		1,80,000	-	-	-	-	Rs. 300 Per Meter	1,80,000
Garland drain 500 Mtrs		1,50,000	-	-	-	-	Rs. 300 Per Meter	1,50,000
TOTAL								**4,30,000**

13.0 MINERAL CONSERVATION AND DEVELOPMENT:

The mining plan proposed has fully covered all the aspects of Granite Conservation and development rules 1999, with a future plan to extend the proposed working of the quarry to the maximum possible workable depth of the deposit. Extreme care is taken to ensure proper supervision of quality control of the Granite dimensional stone aimed at the recovery of the maximum saleable quality and quantity of granite dimensional stones suitable for full utilization of the consumers.

Care is been taken for each process just to safeguard the material quarried in an economical and efficient manner by adopting systematic and scientific quarrying with the consultation and supervision of well experienced quarry persons.

14.0 STATUTORY COMPLIANCE TO BE FOLLOWED DURING THE COURSE & CLOSURE OF QUARRY LEASE:

The company should ensure to carryout that the quarry operation in accordance to the Approved scheme of Mining and as per TNMMCR, 1959, GCDR, 1999, Mine Act 1952 & as per MMR 1961.

The conditions in the nutshell as follows.

1. The quarry operation should be fenced with the barbed wire and three tier Greenbelt has to be carried out all along the boundary barriers and should be maintained till the end of the life of the mine.
2. The quarry operation should be carried out as per the regulation 3,4,5 of MMR 1961 as prescribed in the form I, II, III, IVA, IVB, IVC, V, VI.
3. Statutory competent qualified person as per regulation 39, 40, 41, 42, 43, 44, 47, 48 of MMR 1961 should be appointed to carryout day today quarrying operation in scientific & systematic manner.
4. Blasting has to be carried out during the designated timing as per the approved mining plan with sufficient warning all around the mines by only the Qualified blaster as indicated in the regulations 153, 154, 156, 160, 166, 167, 169 of MMR 1961.
5. During the Quarrying operations the benches has to be maintained and haul roads should be maintained as per the regulations 60, 61, 62, 65, 66, 95, 96, 105, 106, 110, 115, 118, 124 of MMR 1961. When the quarry attains 30m depth from the general ground profile a slope stability study should be carried out with competent authorized agencies only.
6. All PPE should be provided to the employees and periodic Medical Examination has to be carried out as per the regulation 124, 127, 177, 178, 181, 183, 185, 188, 190, 191A of MMR 1961.
7. The company should submit the quarterly, half yearly & Annual returns to the concerned Authorities as per regulation 3, 4, 5, 9 &10 MMR 1961 on time and HYCR self-compliance reports should be submitted to the State Pollution control Board.
8. Periodical Hydrology & Hydrogeological Studies has to be carried out during the course of quarrying operation. if any significant seepages is inferred, the company should carry out a comprehensive Hydrogeological studies to ensure the impact of agricultural activities in and around the mine (Open well & Tube well). If the impact is noticed considerably the quarry had to be continued only after obtaining permission clearances and approval from the competent regulatory authorities.
9. The company should utilize green energy / Solar energy as far as possible for operation of machineries, equipment and illumination in the mine's office, quarry boundaries and avenue roads.
10. The company should ensure all the skilled and semi-skilled workers should be provided trainings in nearby Vocational Training center as per the MVTR 1966 and should be registered in Shram Suvidha portal as directed by the Ministry of labour.
11. The company should carry out the quarrying operation keeping the view of mineral conservation, development and sustainable mining.
12. If the company intends to deviate from the approved scheme of mining, a modified scheme of mining has to be prepared submitted and approved from the concerned authority before making any significant changes in the mining operations.
13. The company should carry out the mining operation in Eco friendly environment with good hygienic conditions to the employees deployed and maintain good harmony with local communities in the impact zone.
14. The conditions stipulated in the lease deed and Environmental clearance especially the mitigation measures as proposed in EMP to the SEIAA has to be followed during the course of quarrying operation and also to as per the rules of TNMCR 1959, GCDR 1999, Mine act 1952, MMR 1961, Mine Rule 1955, Environment Protection Act 1986 and Indian Explosive act 1884 and Indian Explosive Rules 2008 and all the amendments issued by the concern State and Central Government authorities from time to time.

The provisions of the Mines Act, Rules and Regulations and orders made there under shall be complied with, so that the safety of the mine, machinery and person will be well protected. Permission, relaxation or exemption wherever required for the safe and scientific quarrying of the deposit will be obtained from the Department of Mines Safety, Chennai. As per amendment notification in the EIA notification 2006 is given by Ministry of Environment, Forest and Climate Change vide S.O.1807(E), dated:12.04.2022, the validity of environmental clearance is through out end of lease period. Any violation pointed out by the inspecting authorities shall be rectified and modified after scrutiny comments as per the guidelines of the Concerned Department and Authorities.

This Mining Plan and mine design is prepared based on the requirement and full consultation by the applicant to me. If there is any change in the production schedule, change of technology, change in product mix during the course of operations, the applicant is advice to prepare a modified mining plan and get approval by the concerned authority for subsequent clearance and approval. The same will be monitored by the inspecting authority of Department of Geology and mining and other Concerned Departments under Rule 25 and sub rule (5)(d) in Rule 36 of Tamil Nadu Minor Mineral Concession Rules, 1959.

I hereby ensure that the information provided is correct to best of my knowledge and experience, some of the information contained in this report has been provided by external sources and by the applicant and is presented as the form as submitted by the applicant. The information is not intended to serve as legal advice related to the individual situation. I do not owe and specifically disclaim any liability resulting from the use during the course of quarrying operations after the approval of Mining plan or grant of lease. The document may be scrutinized by the competent authority before approval.

Prepared by

B. Vengadagiri, M.Sc.,
Qualified Person

Place: Salem
Date: 18.03.2026.